Process Capability Improvement using Cp/Cpk in Ball Attach Process using Minitab

1. Problem Statement

In this case study, I consider a scenario where variation in solder ball diameter is observed during the ball attach process in semiconductor packaging.

This variation can lead to:

- Open/short defects
- Reliability issues
- Reduced process yield

The objective of this study is to:

a. Evaluate the process capability using Cp/Cpk

b. Identify whether the process is stable and capable

c. Analyze the impact of variation on process performance

d. Propose process improvements to reduce variation and enhance yield

Note:
Variation in ball diameter occurs due to multiple process factors such as temperature variation during reflow, solder volume differences, surface conditions, and material behavior. Since ball attach is a thermal and physical process, perfect uniformity is not achievable, and the goal is to control variation within specification limits. These solder balls are primarily used in BGA packages as external interconnects, while in flip-chip technology, smaller micro-bumps are used for die-to-substrate connection

2. Creating and Analyzing the Dataset (Data-Driven Approach)

Before performing detailed process analysis, I approached the problem from a data perspective, as variation in semiconductor manufacturing is always monitored using production data.

Since real factory data is not available, I created a simulated production-like dataset representing ball diameter measurements from the ball attachment process.

How the Dataset Was Created

In semiconductor manufacturing, process data is typically collected across multiple units processed under similar conditions.

Based on this, I constructed a dataset with the following characteristics:

- Multiple samples representing ball diameter measurements
- Defined specification limits:
    - Lower Spec Limit (LSL) = 95 µm
    - Upper Spec Limit (USL) = 105 µm
- The initial dataset shows higher variation (unstable process condition)
- Improved dataset showing reduced variation (after process optimization)

Dataset Description

Two datasets were considered:

Initial Dataset (Before Improvement)

- Represents process with high variation
- Leads to low process capability

Improved Dataset (After Optimization)

- Represents controlled process
- Reduced variation and improved capability

3. Data Visualization and Initial Observations

To understand the behavior of the ball attach process under initial conditions, statistical analysis was performed using Minitab.

The objective of this analysis is to:

- Evaluate process stability
- Identify variation patterns
- Understand baseline process performance before optimization

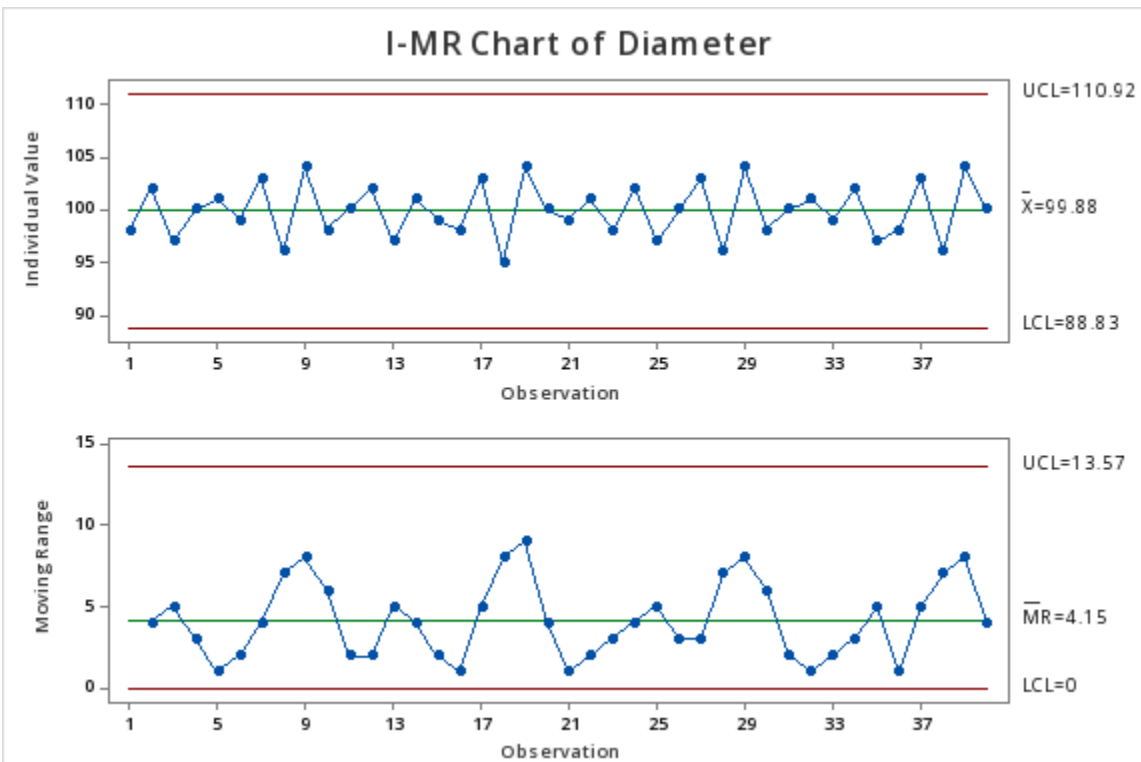
3. 1 Dataset Table:

Observations	Diameter
1	98
2	102
3	97
4	100
5	101
6	99
7	103
8	96
9	104
10	98
11	100
12	102
13	97
14	101
15	99
16	98
17	103
18	95
19	104
20	100
21	99
22	101
23	98
24	102
25	97
26	100
27	103
28	96
29	104
30	98
31	100
32	101
33	99
34	102
35	97
36	98
37	103
38	96
39	104
40	100

Description:

- The dataset represents the **initial state of the process**
- Measurements are taken from multiple units processed under similar conditions
- Specification limits considered:
    - Lower Spec Limit (LSL) = 95 µm
    - Upper Spec Limit (USL) = 105 µm

3.2 Control Chart (I-MR Chart)



Observation:

- All data points lie within the control limits (UCL and LCL)

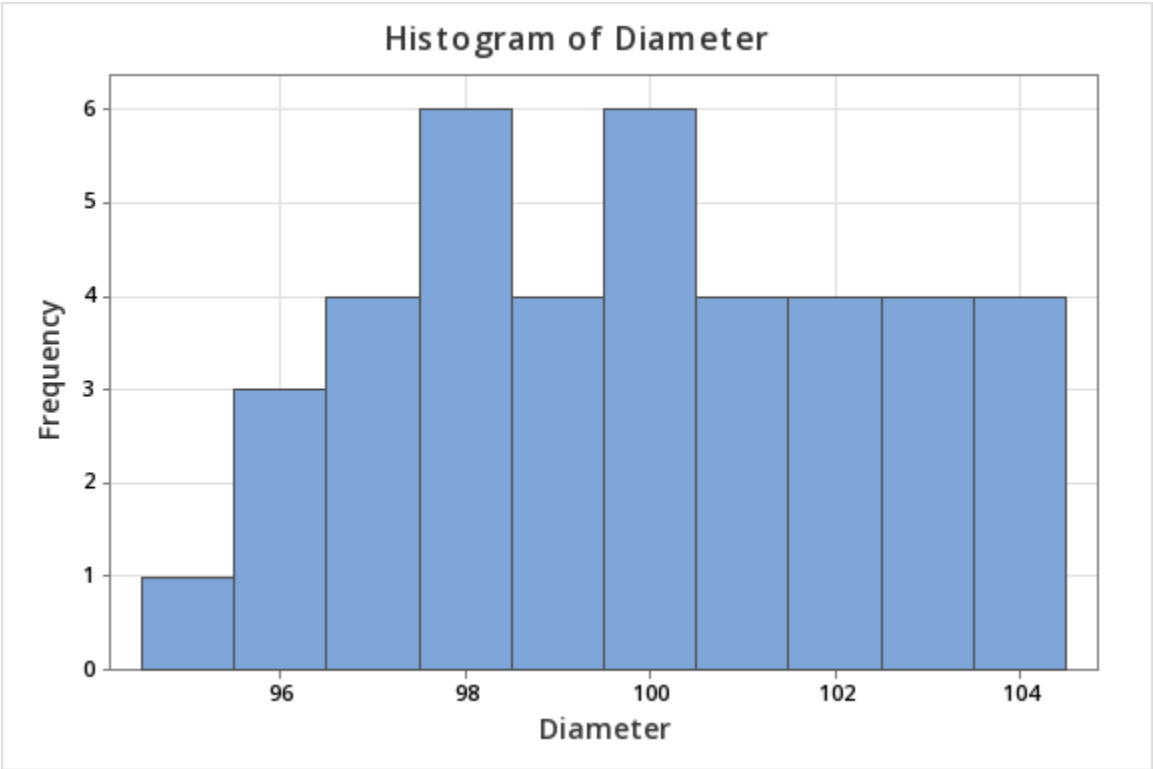
- No out-of-control points or abnormal patterns are observed
- The process appears statistically stable

Interpretation:

The control chart indicates that the process variation is due to common causes rather than sudden disturbances.

However, even though the process is stable, variation still exists within the process.

3.3 Histogram Analysis



Observation:

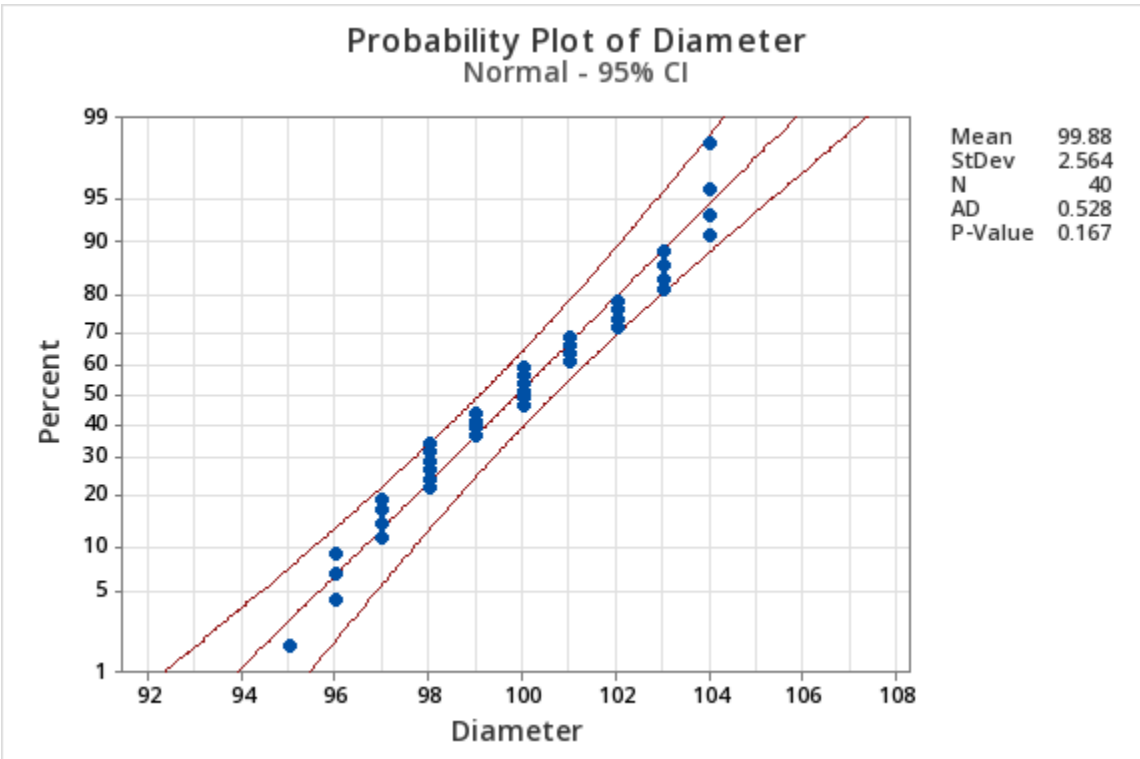
- The histogram shows a relatively wide distribution of data
- The data is spread across the specification range

- The distribution is approximately bell-shaped

Interpretation:

The wider spread indicates high process variation, which may affect process capability and lead to potential defects.

3.4 Probability Plot (Normality Check)



Observation:

- Data points approximately follow a straight line
- P-value indicates that the data follows a normal distribution

Interpretation:

The data is approximately normally distributed, which validates the use of process capability analysis (Cp/Cpk).

Key Insight from Initial Analysis

From the above analysis, it is evident that:

- The process is statistically stable (no special cause variation)
- However, the process exhibits significant variation
- This variation may lead to poor process capability and affect yield

4. Process Capability Analysis

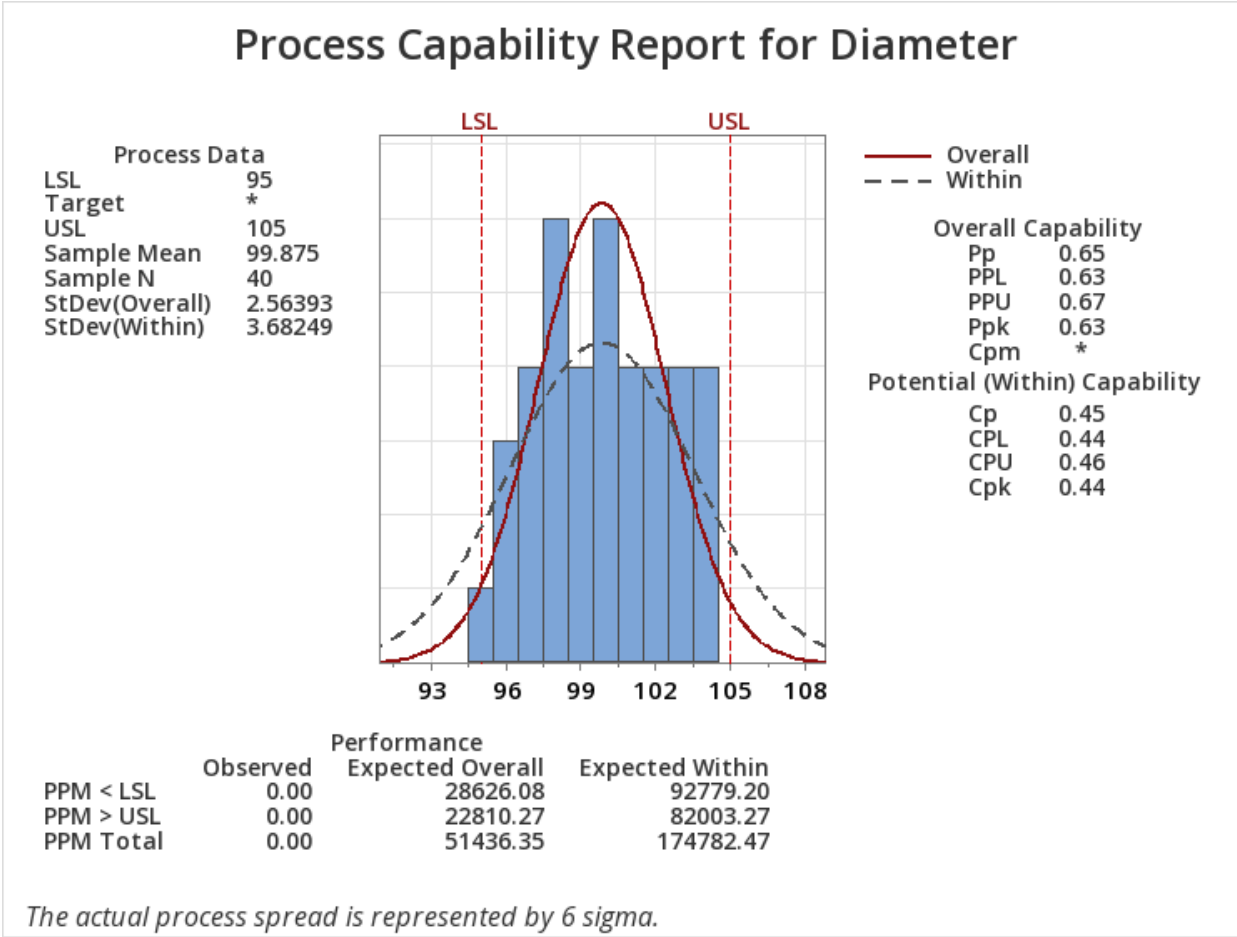
After evaluating the process stability and variation through graphical analysis, the next step is to assess whether the process meets the required specification limits.

Process capability analysis was performed using Minitab to evaluate how well the process can produce outputs within the defined specification range.

4.1 Methodology

The capability analysis was carried out using the following parameters:

- Lower Specification Limit (LSL) = 95 µm
- Upper Specification Limit (USL) = 105 µm
- Data used: Diameter (initial process dataset)



4.2 Results

The capability analysis provides the following key metrics:

- Cp ≈ 0.45
- Cpk ≈ 0.44

4.3 Interpretation

Process Capability (Cp):

- Cp represents the potential capability of the process
- Cp ≈ 0.45 indicates that the process spread is significantly larger than the specification limits

👉 This means:

The process has high variation and cannot consistently meet specifications

Process Performance (Cpk):

- Cpk represents the actual performance considering process centering
- Cpk ≈ 0.44 indicates that the process is not capable

👉 Interpretation:

The process is unable to produce outputs consistently within the specification limits

Important Insight

Even though:

- The process is statistically stable (from control chart)

It is still: Not capable due to high variation

Engineering Conclusion

From the capability analysis, it is evident that:

- The process exhibits excessive variation
- The current process cannot reliably meet specification limits
- There is a need for process optimization to reduce variation

Possible Causes of Variation

The observed variation may be attributed to:

- Temperature fluctuations during reflow
- Variations in solder volume

- Machine or process parameter instability
- Material and surface condition differences

5. Process Improvement and Validation

Based on the capability analysis, it was observed that the process exhibited high variation and was not capable of meeting the required specifications.

To address this issue, process optimization was considered with the objective of reducing variation and improving process performance.

5.1 Improvement Approach

The improvement focuses on reducing variation in the process by controlling key process parameters such as:

- Temperature uniformity during reflow
- Consistency in solder volume
- Machine calibration and alignment
- Process parameter stabilization

These actions aim to:

- Reduce process variation
- Improve consistency
- Enhance overall yield

5.2 Improved Dataset

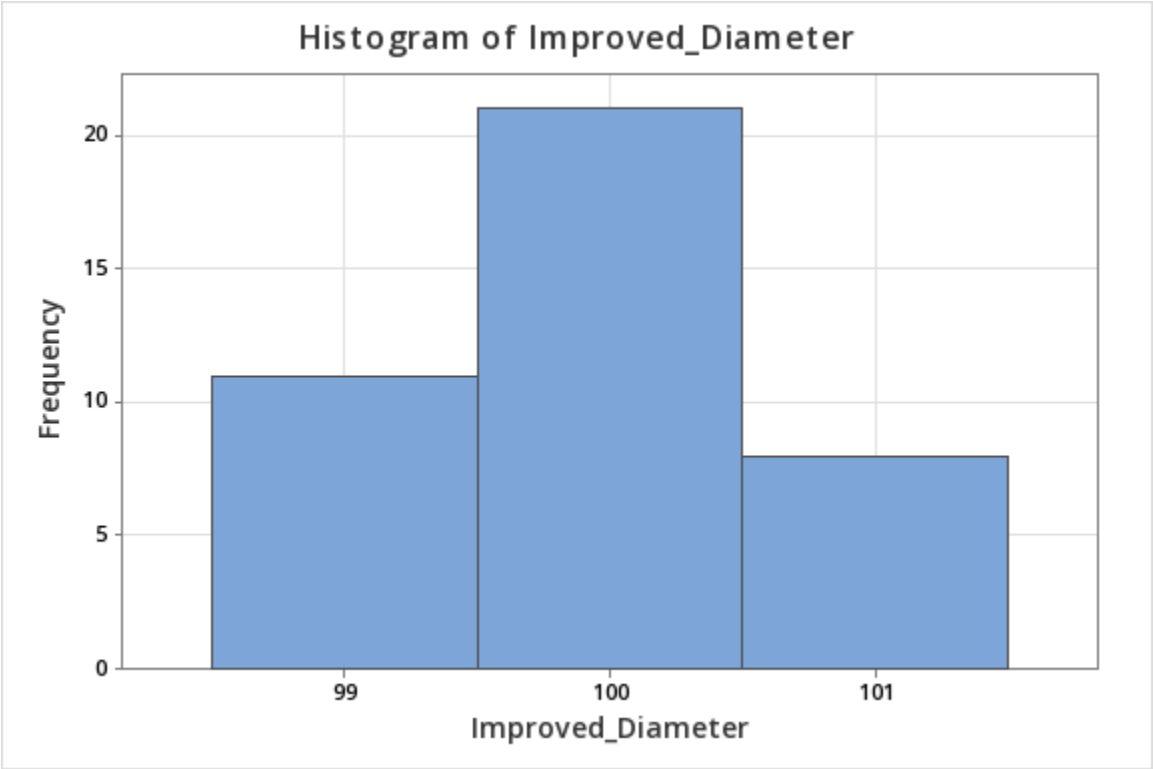
A new dataset (Improved_Diameter) was generated to represent the process after optimization.

This dataset reflects:

- Reduced variation
- More consistent process behavior
- Better centering around the target value

Observations	Diameter	Improved_Diameter
1	98	99
2	102	100
3	97	100
4	100	101
5	101	99
6	99	100
7	103	101
8	96	100
9	104	99
10	98	100
11	100	100
12	102	101
13	97	99
14	101	100
15	99	100
16	98	99
17	103	101
18	95	100
19	104	100
20	100	99
21	99	100
22	101	100
23	98	99
24	102	101
25	97	100
26	100	100
27	103	101
28	96	99
29	104	100
30	98	100
31	100	99
32	101	101
33	99	100
34	102	100
35	97	99
36	98	100
37	103	100
38	96	99
39	104	101
40	100	100

5.3 Histogram Analysis (After Improvement)



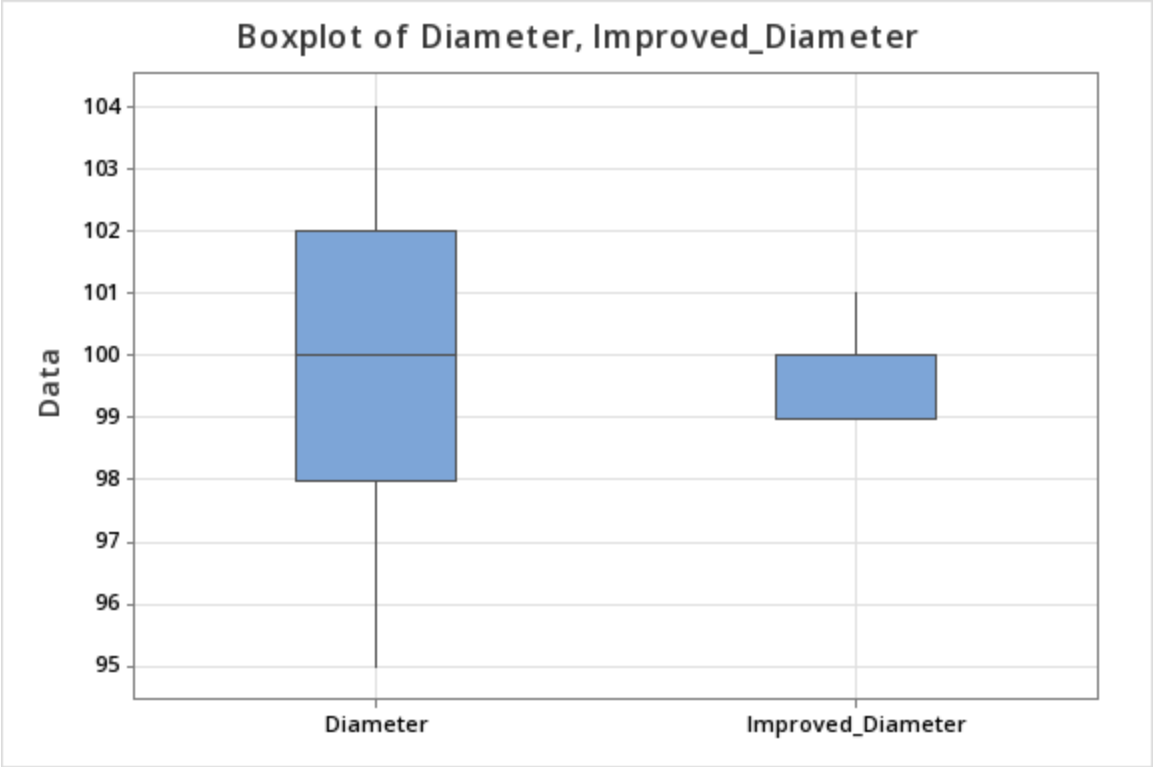
Observation:

- Data is tightly clustered around the mean
- Distribution is narrow compared to initial dataset
- Variation is significantly reduced

Interpretation:

The reduction in spread indicates improved process control and consistency

5.4 Boxplot Comparison (Before vs After)



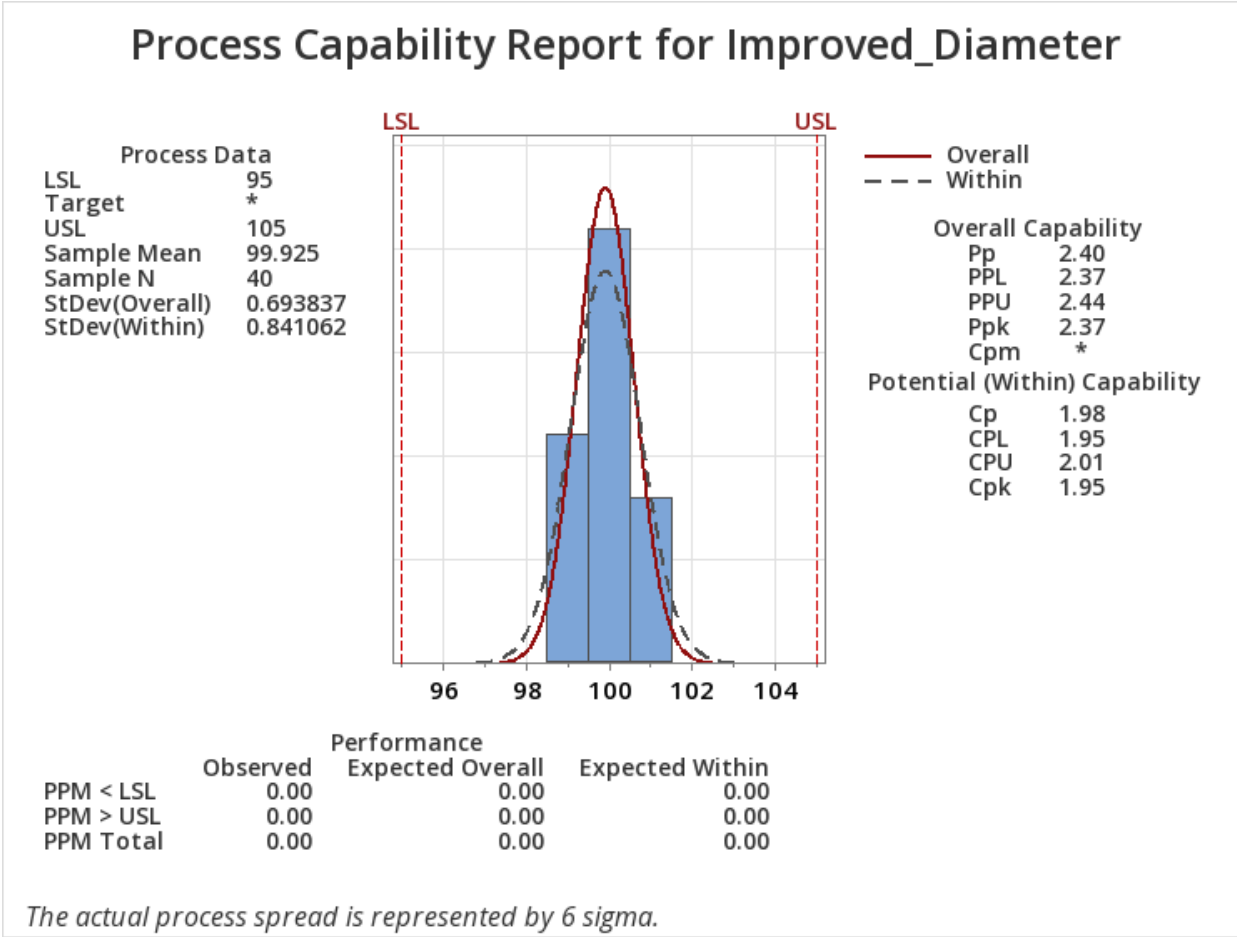
Observation:

- Initial dataset shows larger spread
- Improved dataset shows significantly reduced variation
- Data is tightly clustered after improvement

Interpretation:

The boxplot clearly demonstrates a significant reduction in variation after process optimization

5.5 Process Capability (After Improvement)



Results:

- Ppk ≈ 2.37
- (Cp/Cpk not calculated due to very low variation)

🔍

5.6 Interpretation

- The process capability has improved significantly
- The process is now highly capable and consistent
- Variation has been reduced to a very low level

Meaning: The process can reliably meet specification limits

Final Comparison

Metric	Before	After
Variation	High	Low
Cpk	≈ 0.44	-
Ppk	-	≈ 2.37
Process Capability	Not capable	Highly capable

5.8 Engineering Conclusion

From the analysis, it can be concluded that:

- The initial process was stable but not capable due to high variation
- Process optimization significantly reduced variation
- The improved process demonstrates high capability and consistency
- This leads to improved yield and reduced defect risk

FINAL INSIGHT (VERY IMPORTANT — Add this)

This study highlights that reducing process variation is critical for improving process capability and achieving reliable manufacturing performance in semiconductor packaging.